

CoRoT-2b

R-1381

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_190920 · created 2026-08-01T01:57:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458746.6712 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.014
Transit depth (Rp/Rs)^2	2.25 +/- 0.43 [%]
Orbital Inclination (inc)	89.92 +/- 1.01
Airmass coefficient 1 (a1)	0.98 +/- 0.012
Airmass coefficient 2 (a2)	0.015 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#2 - [612, 258]
Optimal Aperture	3.79
Optimal Annulus	16.21
Transit Duration (day)	0.0952 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.014 vs 0.1667 (deviation 1.19σ)
Duration fit vs published	0.0952 d vs 0.095 d (deviation 0.14σ)
Mid-time O–C	2.1 min
Depth SNR	10.7
Residual scatter	1.26 %

Light curve

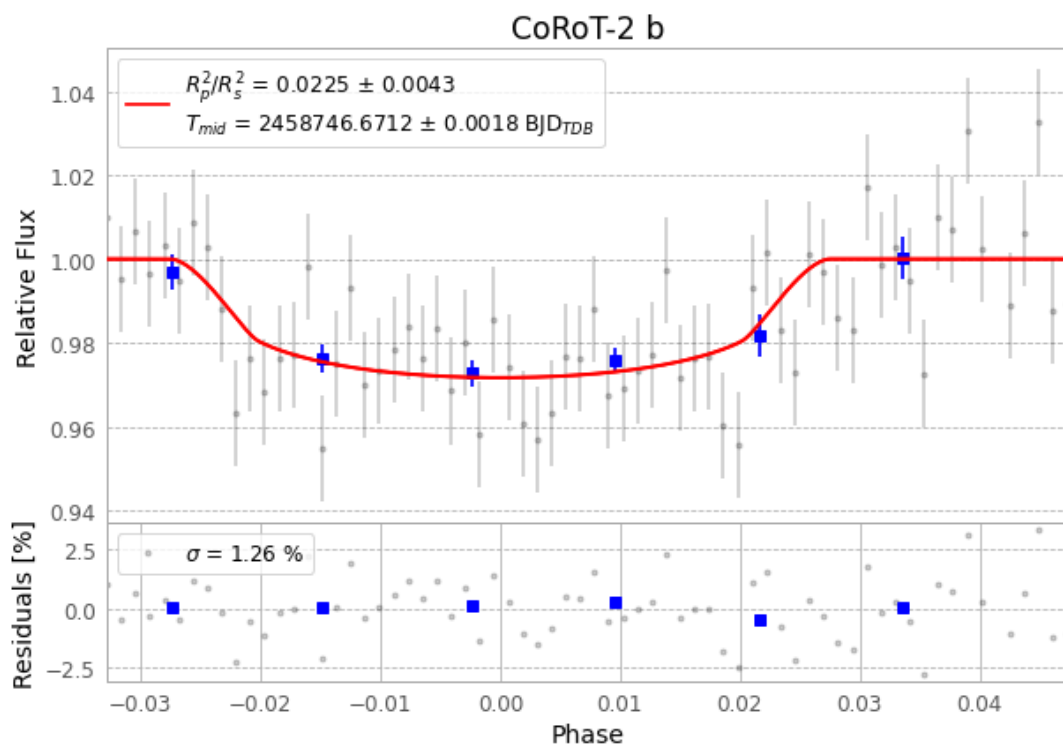


Figure 1 — FinalLightCurve_CoRoT-2 b_2019-09-19.png (SHA-256 18a58135ccd2d43c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [142, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6787768b68b20b07...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), CoRoT-2190920023912.FITS → CoRoT-2190920060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-190920.log (8b2e5109c2fa...), FinalLightCurve_CoRoT-2 b_2019-09-19.png (18a58135ccd2...), FinalLightCurve_CoRoT-2 b_2019-09-19.csv (91add49ff9c1...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:57Z.